# Practice workbook · Describe transcription and translation in human cells

Describing how a cell makes protein from a gene, and applying the sequence to switched-off genes, low amino acid supply or viral proteins.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-PROTEIN-SYNTHESIS

# 1. Question 8 · part a.ii · 5 marks

June 2023 Paper 2R

Worked-example retry · Attempt part a.ii only.

**8** (a) (i) Describe the structure of a virus.

**(2)**

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

**(5)**

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# 1. Question 8 · part a.ii · 5 marks

June 2023 Paper 2R

Worked-example retry · Attempt part a.ii only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

**(1)**

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

**(2)**

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percentage decrease = ..... %

**(Total for Question 8 = 10 marks)**

## 1. Question 8 · part a.ii · 5 marks · Working and answer

Working and answer

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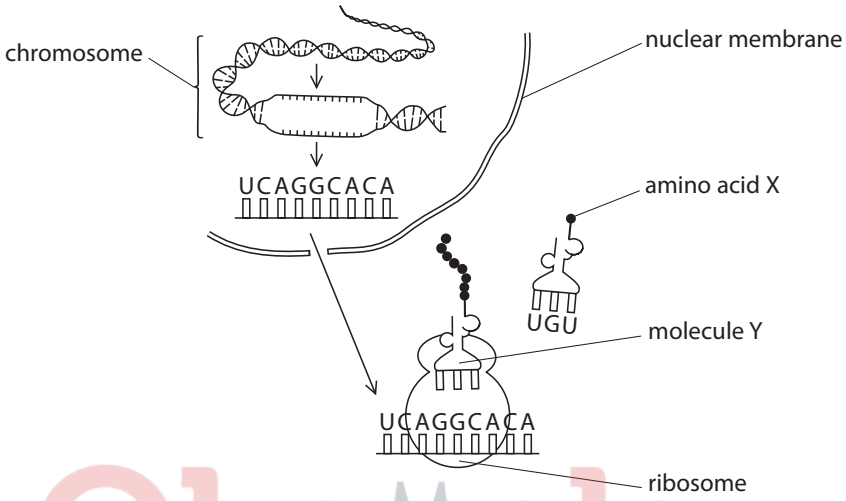
## 2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

**6** Protein synthesis has a number of stages.

The diagram shows some of the stages.




The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

| Base triplet of DNA | Amino acid |
|---|---|
| AGT | serine |
| CCG | glycine |
| TGT | threonine |
| GTA | histidine |
| CAA | valine |
| TCC | argenine |
| ACA | cysteine |

(a) Describe the function of the nuclear membrane.

**(2)**

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## 2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

**(1)**

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

**(1)**

(d) Describe the process occurring on the ribosome.

**(6)**

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## 2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(e) Describe how a mutation will affect a chromosome.

**(2)**

**(Total for Question 6 = 12 marks)**

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Working and answer

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## 3. Question 9 · part a.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

**9** (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

**(3)**

(ii) Explain why people with PKU are advised to follow a low-protein diet.

**(3)**

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

**(3)**

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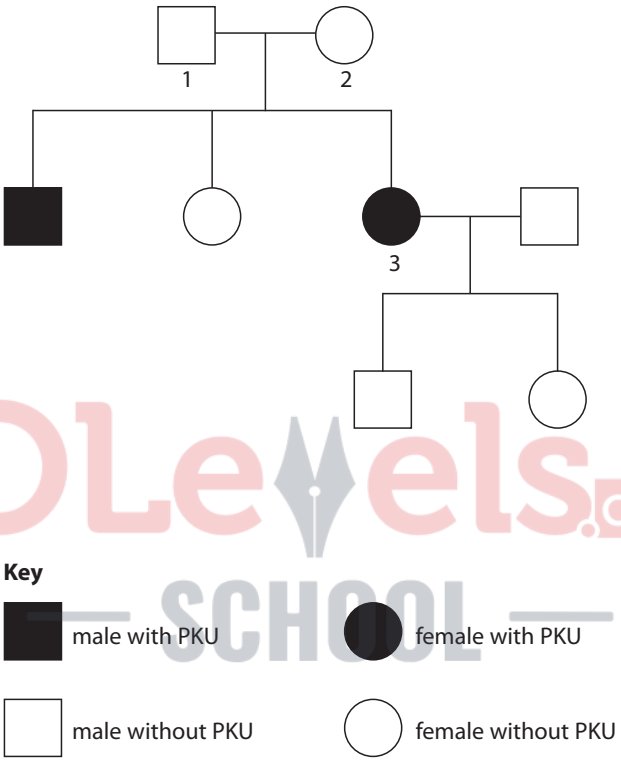
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## 3. Question 9 · part a.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) The diagram shows how PKU is inherited in a family.




(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

**(1)**

(ii) Determine the probability that person 2 has one allele for PKU.

**(1)**

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## 3. Question 9 · part a.iii · 3 marks · Working and answer

Working and answer

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# 4. Question 5 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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# 4. Question 5 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

**(2)**

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

**(2)**

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

**(4)**

**(Total for Question 5 = 14 marks)**

## 4. Question 5 · part c.i · 3 marks · Working and answer

Working and answer

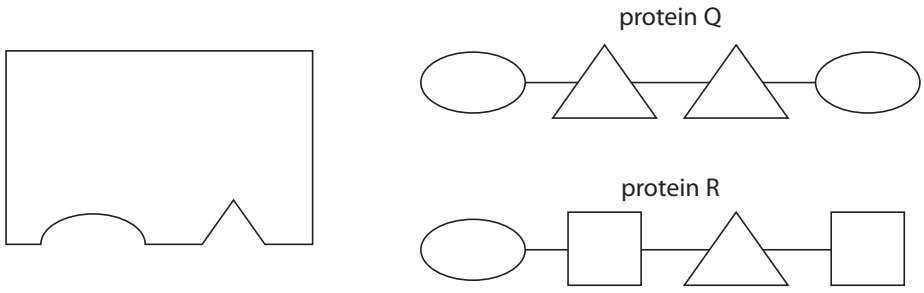
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# 5. Question 2 · part a.iii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | |
|---|---|---|---|
| amino acids | amylases | fatty acids | proteases |
| protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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# 5. Question 2 · part a.iii · 1 mark

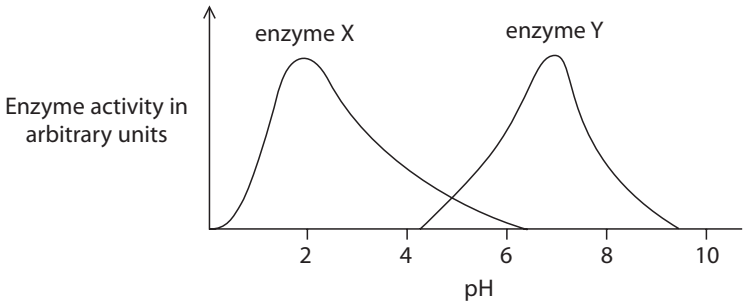
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Further practice · Attempt part a.iii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.




(i) Describe how the activity of enzyme X changes with pH.

**(3)**

(ii) Explain why no activity is shown for enzyme Y at pH2.

**(2)**

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## 5. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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